

LAWS OF MOTION

Important formulas and concepts directly from the NCERT-

➤ State of rest and state of constant velocity are natural states (net force = 0)

❖ Newton's First Law-

- No net force is required to keep the object in same state.
- Object in motion stays in motion or if at rest remains at rest unless acted by an external force.

❖ Inertia –

- Property which oppose cause of change in state of a body.
- Inertia $\propto$ mass
- Types of inertia-
 - 1) Inertia of rest
 - 2) Inertia of uniform motion

❖ Normal reaction-

- Act due to contact.
- Act always perpendicular to contact surface.
- Always in pair.

❖ Tension-

- Away from contact point.
- Along the length of string.
- In case of ideal string, tension same at all point.

❖ Spring force-

- Always act towards the mean position.
- Spring force, $F_s = kx$

❖ Newton's Second law of motion-

$$\text{➤ } F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

If mass is constant then,

$$F = ma \quad (a = \text{acceleration})$$

➤ Force is rate of change of momentum,

$$F = \frac{dP}{dt}$$

➤ Impulse is change in momentum equals to force $\times$ time.

❖ Newton's Third law of motion-

➤ For every action, there is an equal and opposite reaction.

➤ Action reaction pair acts upon two different bodies, act at same time and both have equal magnitude and opposite direction.

❖ Rocket-

$$\text{➤ Upthrust} = v \frac{dm}{dt}$$

$$\text{➤ Initial net force, } v \frac{dm}{dt} - m_0 g$$

$$\text{➤ Initial acceleration } (a_0) = (v \frac{dm}{dt}) / m_0 - g \quad (m_0 = \text{initial mass})$$

$$\text{➤ At time } t, \text{ mass } (m_t) = m_0 - (dm/dt)t$$

➤ At time t , in acceleration and net force, we put m_t instead of m_0 .

❖ Gun – bullet-

➤ Magnitude of momentum of gun and bullet are same but opposite in direction.

$$M_b v_b = -M_G v_G$$

$$\text{Recoil speed, } v_G = M_b v_b / M_G$$

➤ Magnitude of momentum is same,

$$\text{Kinetic energy } \propto 1/m \quad (K = P^2/2m)$$

$$K_G/K_b = m_b/m_G$$

➤ **Force on 'N' bullets = $nMbv_b$ = force on gun**

$n = N/t$ (bullets per second)

pu

➤ **If any body is moving upward,**

Normal rxⁿ , $N = m(g+a)$

If any body is moving downward,

$N = m(g-a)$

❖ **Pulley system-**

➤ For a pulley system of two masses,

Common acceleration, $a_c = (m_2 - m_1)g / (m_1 + m_2)$

❖ **For a block to remain stationary on an inclined plane,**

○ **Acceleration of inclined plane, $a = g \tan \beta$ (β = angle made by inclined plane with horizontal)**

If $a < g \tan \beta$ block will slide up

If $a > g \tan \beta$ block will slide down

○ **Force applied by inclined plane on block, $N = mg / \cos \beta$**

❖ **Friction –**

➤ **It is a contact force.**

○ **Limiting friction(f_s)_{lim} – minimum force required to move an object and maximum force applied to stop an object.**

○ **Static friction(f_s) – it is self adjusting according to requirement. It doesn't have fixed value .**

➤ **Necessary conditions for friction-**

1) Contact (normal reaction)

2) Interlocking (coefficient of friction, μ)

3) Tendency of relative motion

$$f_s \leq (f_s)_{\text{lim}} = \mu_s N \quad (\mu_s = \text{coefficient of static friction})$$

$$f_k = \mu_k N \quad (\mu_k = \text{coefficient of kinetic friction})$$

➤ **Net contact force,**

$$F_c = \sqrt{(\text{Normal reaction})^2 + (\text{friction})^2}$$

➤ **Angle of friction-**

- Angle b/w normal reaction and net contact force (when limiting friction acts means about to move).

- $\tan \theta = \mu$

➤ **angle of repose-**

- maximum angle of inclination such that block remains stationary on inclined plane.
- $\tan \beta = \mu$

❖ **friction on inclined plane-**

- **case 1 - $\tan \alpha = \mu$,** block is just about to slide ($\alpha = \text{angle of inclined plane with horizontal}$)
- **case 2 - $\tan \alpha < \mu$,** block remains at rest (**static friction = $mg \sin \alpha$**)
- **case 3 - $\tan \alpha > \mu$,** block will move with acceleration , **$a = g \sin \alpha - \mu g \cos \alpha$**

❖ **death well-**

➤ $\omega_{\min} = \sqrt{g/\mu R}$

❖ **motion of car on rough horizontal road-**

- maximum safe speed of car,

$$v_{\max} = \sqrt{\mu R g}$$

❖ **motion of car on smooth banked road-**

- safe speed of car,

$$v = \sqrt{R g \tan \theta}$$

❖ **bending of cyclist –**

- angle by which cyclist bend with vertical,

$$\tan\theta = v^2/Rg$$

- speed of cyclist,

$$v = \sqrt{Rg \tan\theta}$$

❖ **time period of conical pendulum-**

$$T = 2\pi\sqrt{l\cos\theta/g}$$

SUMMARY

1. Aristotle's view that a force is necessary to keep a body in uniform motion is wrong. A force is necessary in practice to counter the opposing force of friction.
2. Galileo extrapolated simple observations on motion of bodies on inclined planes, and arrived at the law of inertia. Newton's first law of motion is the same law rephrased thus: "*Everybody continues to be in its state of rest or of uniform motion in a straight line, unless compelled by some external force to act otherwise*". In simple terms, the First Law is "**If external force on a body is zero, its acceleration is zero**".
3. Momentum (**p**) of a body is the product of its mass (*m*) and velocity (**v**) :

$$\mathbf{p} = m \mathbf{v}$$

4. Newton's second law of motion :
The rate of change of momentum of a body is proportional to the applied force and takes place in the direction in which the force acts. Thus

$$\mathbf{F} = k \frac{d\mathbf{p}}{dt} = k m \mathbf{a}$$

where **F** is the net external force on the body and **a** its acceleration. We set the constant of proportionality $k = 1$ in SI units. Then

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\mathbf{a}$$

The SI unit of force is newton : $1 \text{ N} = 1 \text{ kg m s}^{-2}$.

- (a) The second law is consistent with the First Law (**F** = 0 implies **a** = 0)
 - (b) It is a vector equation
 - (c) It is applicable to a particle, and also to a body or a system of particles, provided **F** is the total external force on the system and **a** is the acceleration of the system as a whole.
 - (d) **F** at a point at a certain instant determines **a** at the same point at that instant. That is the Second Law is a local law; **a** at an instant does not depend on the history of motion.
4. Impulse is the product of force and time which equals change in momentum.
 The notion of impulse is useful when a large force acts for a short time to produce a measurable change in momentum. Since the time of action of the force is very short, one can assume that there is no appreciable change in the position of the body during the action of the impulsive force.
 6. Newton's third law of motion:
To every action, there is always an equal and opposite reaction

In simple terms, the law can be stated thus :

Forces in nature always occur between pairs of bodies. Force on a body A by body B is equal and opposite to the force on the body B by A.

Action and reaction forces are simultaneous forces. There is no cause-effect relation between action and reaction. Any of the two mutual forces can be called action and the other reaction. Action and reaction act on different bodies and so they cannot be cancelled out. The internal action and reaction forces between different parts of a body do, however, sum to zero.

7. *Law of Conservation of Momentum*

The total momentum of an isolated system of particles is conserved. The law follows from the second and third law of motion.

8. *Friction*

Frictional force opposes (impending or actual) relative motion between two surfaces in contact. It is the component of the contact force along the common tangent to the surface in contact. Static friction f_s opposes impending relative motion; kinetic friction f_k opposes actual relative motion. They are independent of the area of contact and satisfy the following approximate laws :

$$f_s \leq (f_s)_{\max} = \mu_s R$$

$$f_k = \mu_k R$$

μ_s (co-efficient of static friction) and μ_k (co-efficient of kinetic friction) are constants characteristic of the pair of surfaces in contact. It is found experimentally that μ_k is less than μ_s .

POINTS TO PONDER

1. Force is not always in the direction of motion. Depending on the situation, $\mathbf{F}$ may be along $\mathbf{v}$, opposite to $\mathbf{v}$, normal to $\mathbf{v}$ or may make some other angle with $\mathbf{v}$. In every case, it is parallel to acceleration.
2. If $\mathbf{v} = 0$ at an instant, i.e. if a body is momentarily at rest, it does not mean that force or acceleration are necessarily zero at that instant. For example, when a ball thrown upward reaches its maximum height, $\mathbf{v} = 0$ but the force continues to be its weight mg and the acceleration is not zero but g .
3. Force on a body at a given time is determined by the situation at the location of the body at that time. Force is not 'carried' by the body from its earlier history of motion. The moment after a stone is released out of an accelerated train, there is no horizontal force (or acceleration) on the stone, if the effects of the surrounding air are neglected. The stone then has only the vertical force of gravity.
4. In the second law of motion $\mathbf{F} = m \mathbf{a}$, $\mathbf{F}$ stands for the net force due to all material agencies external to the body. $\mathbf{a}$ is the effect of the force. $m\mathbf{a}$ should not be regarded as yet another force, besides $\mathbf{F}$.

5. The centripetal force should not be regarded as yet another kind of force. It is simply a name given to the force that provides inward radial acceleration to a body in circular motion. We should always look for some material force like tension, gravitational force, electrical force, friction, etc as the centripetal force in any circular motion.
6. Static friction is a self-adjusting force up to its limit $\mu_s N$ ($f_s \leq \mu_s N$). Do not put $f_s = \mu_s N$ without being sure that the maximum value of static friction is coming into play.
7. The familiar equation $mg = R$ for a body on a table is true only if the body is in equilibrium. The two forces mg and R can be different (e.g. a body in an accelerated lift). The equality of mg and R has no connection with the third law.
8. The terms 'action' and 'reaction' in the third Law of Motion simply stand for simultaneous mutual forces between a pair of bodies. Unlike their meaning in ordinary language, action does not precede or cause reaction. Action and reaction act on different bodies.
9. The different terms like 'friction', 'normal reaction', 'tension', 'air resistance', 'viscous drag', 'thrust', 'buoyancy', 'weight', 'centripetal force' all stand for 'force' in different contexts. For clarity, every force and its equivalent terms encountered in mechanics should be reduced to the phrase 'force on A by B'.
10. For applying the second law of motion, there is no conceptual distinction between inanimate and animate objects. An animate object such as a human also requires an external force to accelerate. For example, without the external force of friction, we cannot walk on the ground.
11. The objective concept of force in physics should not be confused with the subjective concept of the 'feeling of force'. On a merry-go-around, all parts of our body are subject to an inward force, but we have a feeling of being pushed outward – the direction of impending motion.